

Association Between State-Level Public Education Spending and School Vaccination Rates in the United States

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1. Introduction

Childhood vaccination coverage is an important measure of how well children in a population are protected against vaccine-preventable diseases. In this report, we investigate whether the amount each state spends on public education is associated with school vaccination coverage in that state. To address this question, we use a cross-sectional dataset formed by merging school-specific vaccination data from U.S. states with state education spending data. The dataset includes more than 46,000 schools, with over 42,000 reporting at least one vaccination measure, and contains information on overall vaccination coverage, measles, mumps, and rubella (MMR) vaccination coverage, school characteristics, and state-level measures such as public spending on elementary and secondary education and the estimated school-aged population.

2. Methods

2.1. Data sources and study design

This study used a cross-sectional design to examine whether state-level public spending on elementary and secondary education was associated with school-level vaccination rates in the United States. The dataset consisted of school-specific vaccination data from 32 U.S. states merged with 2016 state-level education spending data. The vaccination data were originally collected by reporters at the *Wall Street Journal* from state health departments. Most observations were from the 2018-2019 school year, while 11 states provided data from 2017-2018. State-level spending data came from 2016 public spending estimates based on the U.S. Census Bureau's *Annual Survey of State and Local Government Finances* and other federal and noncensus sources.

2.2. Measures

At the state level, the dataset included vaccination data for the 2017-2018 school year for Colorado, Connecticut, Minnesota, Montana, New Jersey, New York, North Dakota, Pennsylvania, South Dakota, Utah and Washington, while the remaining states contributed data from 2018-2019. The dataset also included the estimated school-aged population and total public spending on elementary and secondary education in 2016, reported in thousands of dollars. At the school level, the dataset included school name, county, school type, and both overall and MMR vaccination rates for individual schools.

2.3. Statistical methods

All analyses were conducted in R (version 4.5.3). An analytic dataset was created by cleaning the merged raw data: selected character codes such as "null", blank strings, and -1 for vaccination outcomes were treated as missing. School-year-location duplicates were handled by firstly removing exact duplicates and then selecting the entry with the most complete vaccination data and if equal, the entry with the largest school enrollment was kept.

To handle missingness in enrollment and vaccination outcomes, multiple imputation ($m=5$) was performed using the Multiple Imputation by Chained Equations (MICE) framework with a Classification and Regression Tree (CART) iterative model using R's *mice* package. The imputation process was guided by a custom predictor matrix to enforce specific relational dependencies: school enrollment was imputed using state-level education spending as the sole predictor. Subsequently, these imputed enrollment values were used in conjunction with state education spending to impute MMR

and overall vaccination rates. On the other hand, state education spending was treated as a complete-case covariate and was not subject to imputation. School type was also retained as a non-imputed categorical variable. School type was categorised into four groups: Public (reference group), Private, Other (combining BOCES, Charter, Kindergarten, and nonpublic schools), and Unknown. Missing school type was not imputed and was instead retained as an explicit “Unknown” category. Exploratory data analysis was first performed to assess the distribution of vaccination rates and variation in state-level education spending. A spending per student variable was created by dividing the total state-level education spending by the total school-aged population of the state; this variable was then standardised to have a mean of 0 and a standard deviation of 1. Descriptive statistics for the predictor variables were summarized.

Separate random intercepts linear mixed models were fitted for overall vaccination rates and MMR vaccination rates to estimate the association between state-level public education spending and school vaccination rates. To account for the hierarchical nesting of schools within states, we included a random intercept for each state. This allowed the model to capture unobserved state-specific factors, such as variations in immunization laws and public health infrastructure, that might influence school-level outcomes. The vaccination rates were log-reflected to correct left skewness and linear regression assumption violations in the QQ plot. For each vaccination rate R , the transformed outcome was defined as

$$Y = \log((\max + 1) - R),$$

where 100 is the maximum possible vaccination rate. Under this transformation, a negative coefficient corresponds to an increase in the original vaccination rate, so to make correct interpretations, the sign has to be reversed.

The models included standardized state education spending per student and school type as fixed effects. The full model was:

$$Y_{ij} = \beta_0 + \beta_1 \text{Spending}_j + \beta_2 \text{Other}_{ij} + \beta_3 \text{Private}_{ij} + \beta_4 \text{Unknown}_{ij} + u_{0j} + \epsilon_{ij} \quad (1)$$

where Y_{ij} is the log-reflected vaccination rate for school i in state j , β_0 is the intercept for public schools at average spending, β_1 is the fixed effect of standardized state education spending per student, β_2 , β_3 , and β_4 are fixed effects for Other, Private, and Unknown school types relative to Public, u_{0j} is the random intercept for state j , and ϵ_{ij} is the residual error.

Sensitivity analyses were conducted by stratifying the data by school type to assess whether the association between public spending and vaccination rates differed across school types. Additional sensitivity analyses stratified the data by vaccination year to assess whether the relationship was consistent across the 2017-2018 and 2018-2019 cohorts. Additional analyses were conducted to assess patterns of missingness and their associations with spending, school type, and state-level clustering.

3. Results

3.1. Sample characteristics

Table 1 presents descriptive statistics by vaccination reporting status. Of the 42,108 schools with at least one observed vaccination rate before imputation, 12,244 reported both MMR and overall rates, 16,058 reported MMR only, and 13,806 reported overall rates only. Schools reporting MMR only had the highest average enrollment (223 students), while schools reporting overall rates only were concentrated in lower-spending states and consisted entirely of schools that were neither public nor private (i.e., charters, kindergartens, nonpublic, etc.). The state-level education spending per student had a mean of \$11,037 and ranged between \$5,885 and \$18,999. The number of unique states in each category reflects that there is at least one school in that state reporting that level of vaccination data.

Table 1: Descriptive Statistics by Vaccination Reporting Status

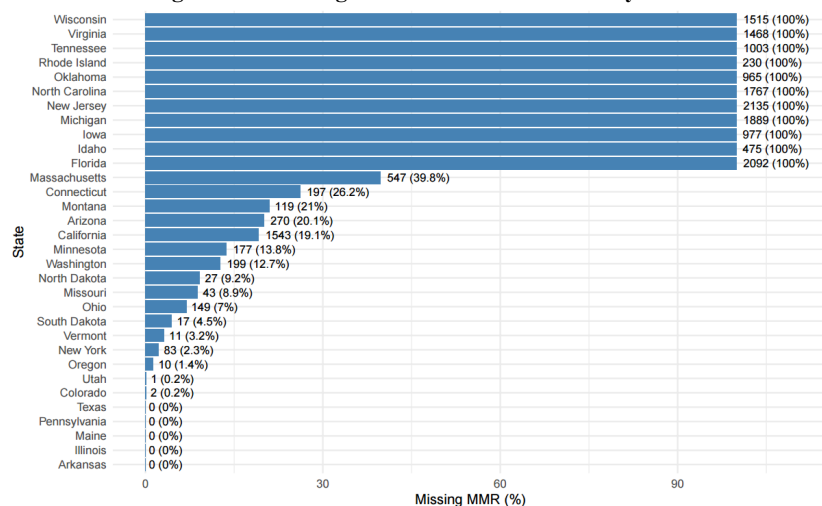
Variable	Full Sample	Overall and MMR	MMR Only	Overall Only
Unique States	32	6	18	11
Avg Schools per State	1315.9 (1270.6)	2040.7 (2265.4)	892.1 (1057.6)	1255.1 (641.7)
State Spending in 1000s (2016)	26893601 (22710345.3)	42609936.1 (28347717.1)	25466490.5 (19382309.5)	14615296.9 (7162176.8)
Total Number of Schools (n)	42108	12244	16058	13806
Enrollment per School	126.1 (163.9)	77.2 (56.6)	222.7 (244.2)	87.5 (92.4)
Public Schools (n)	11729	7232	4497	0
Private Schools (n)	3125	1288	1837	0
Other Schools (n)	27254	3724	9724	13806

Note. The unique states show which states have at least one school that reports either just the MMR, just the overall rate or both, so there exists double counting of states across the three categories.

3.2. Missing data patterns

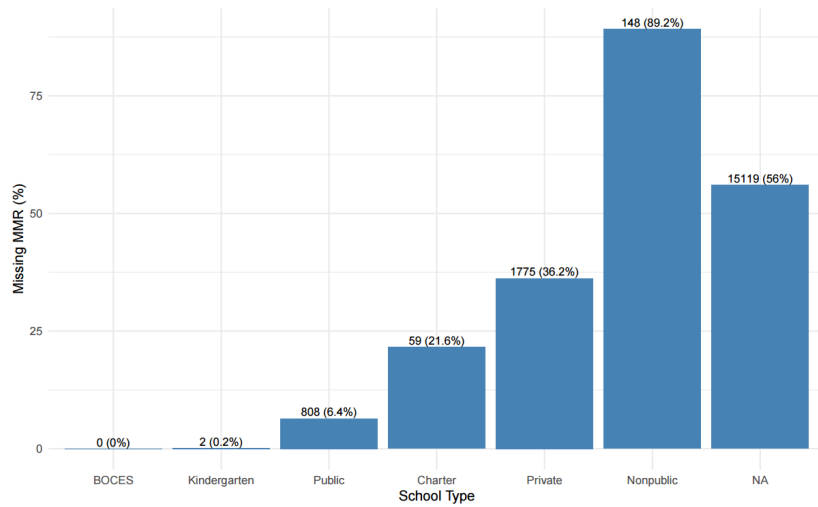
Missingness varied substantially by state and school type. For MMR, 10 states from the 32 in the analysis had fully missing MMR vaccination data in every school. Interestingly, nonpublic schools had the highest rate of missingness, followed by private schools, potentially due to an obligation to report these for public schools. Similar differences by state and school type were observed for overall vaccination rates (Figure 1). Nonpublic schools had higher MMR missing rates (89.2%) than private schools (36.2%) and public schools (6.4%) (Figure 1B). For overall vaccination rates, missingness varied by state (Figure 1C) and school type (Figure 1D).

To assess whether missingness was associated with education spending, standardized spending was compared between schools with observed and missing outcomes. For MMR, schools with missing data were located in lower-spending states on average than schools with observed data (mean scaled spending per student: -0.24 vs. 0.15; $t = 43.46$, $p < 0.001$). For overall vaccination rates, the pattern was reversed: schools with missing data were located in higher-spending per student states on average than schools with observed data (0.33 vs. -0.26; $t = -61.82$, $p < 0.001$). Figures 2A and 2B further illustrate the relationship between state education spending and average observed vaccination rates at the state level.

Figure 1A: Missing MMR Vaccination Rates by State

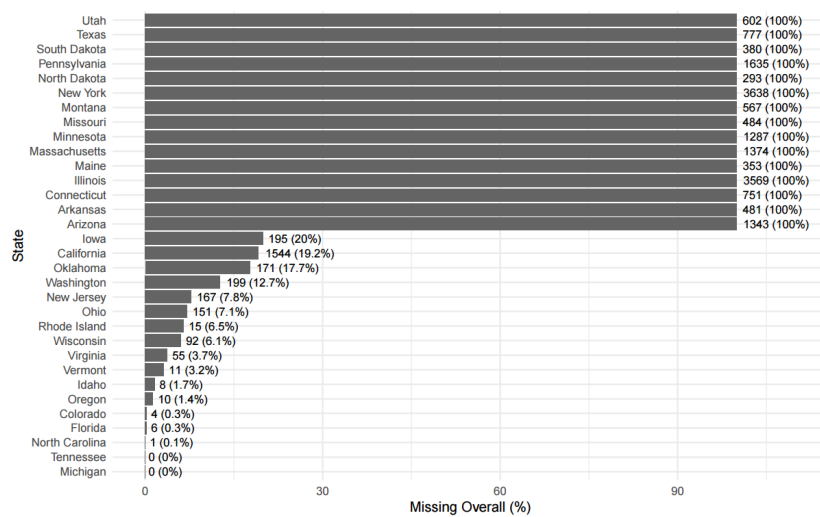
Note. The bar plot only shows percentage of missing MMR rates at the school level for the states with at least one school in the dataset.

Figure 1B: Missing MMR Vaccination Rates by School Type



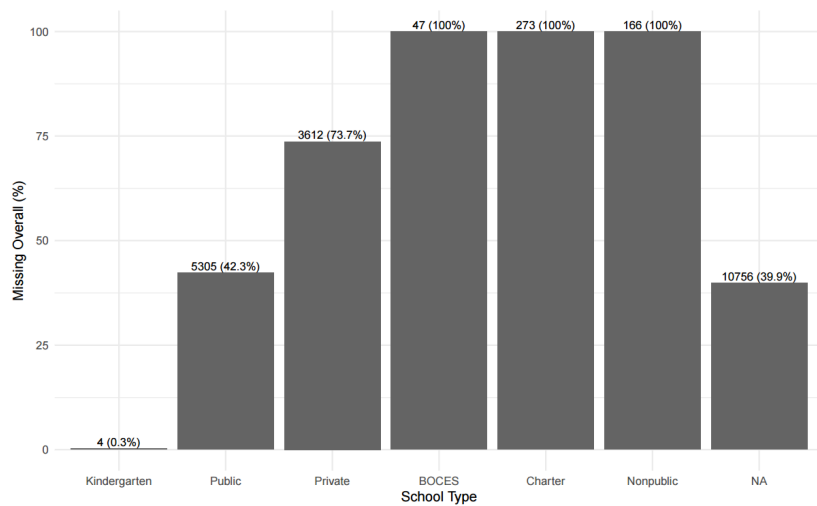
Note. In the analysis, BOCES, Kindergarten, charter and nonpublic were grouped into the other school type. Missing entries for school type were grouped into an “unknown” school type.

Figure 1C: Missing Overall Vaccination Rates by State



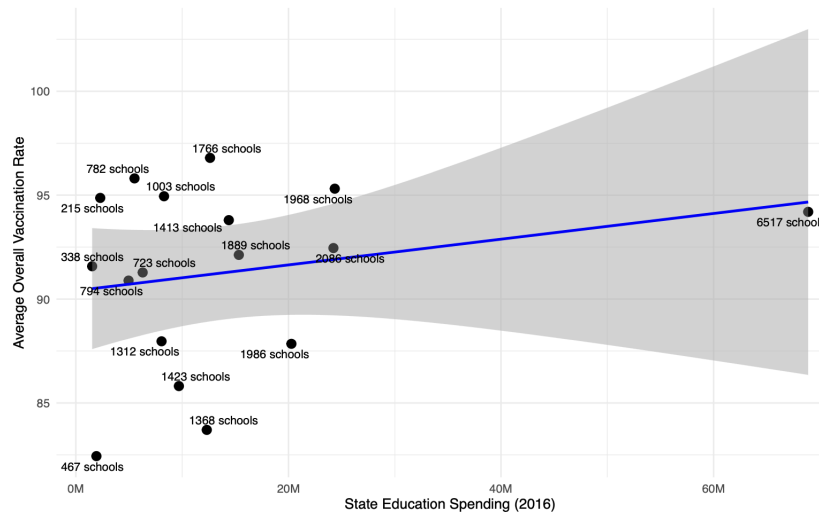
Note. The bar plot only shows percentage of missing overall rates at the school level for the states with at least one school in the dataset.

Figure 1D: Missing Overall Vaccination Rates by School Type



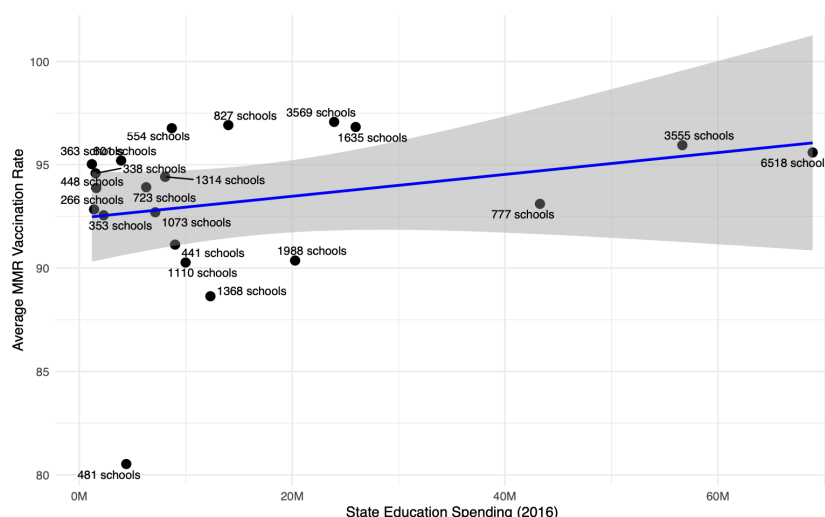
Note. In the analysis, BOCES, Kindergarten, charter and nonpublic were grouped into the other school type. Missing entries for school type were grouped into an “unknown” school type.

Figure 2A: State Education Spending and Average Observed Overall Vaccination Rate (n = 17 states)



Note. The scatterplot shows the relationship between state education spending with average overall vaccination rate with the number of schools in that state.

Figure 2B: State Education Spending and Average Observed MMR Vaccination Rate (n = 21 states)



Note. The scatterplot shows the relationship between state education spending with average MMR vaccination rate with the number of schools in that state.

3.3. MMR vaccination rates

After data cleaning and multiple imputation, the analytic sample included 46,213 schools across 32 states. Before imputation, 42,108 schools reported at least one vaccination rate. In the full analytic sample, the mean MMR vaccination rate was 93.9% (SD = 9.4, range: 1–100), the mean overall vaccination rate was 90.9% (SD = 13.7, range: 1–100), and mean state education spending per student was \$11,037 (SD = \$3,350; range: \$5,885–\$18,999). Table 2 summarizes the mixed model results.

The null model showed that 24.1% of the variance in MMR rates was attributable to between-state differences (ICC = 0.241). After adjustment for spending and school type, the ICC was 0.225. In the adjusted model, state education spending was significantly associated with MMR vaccination rates ($\beta_1 = -0.185$, $p=0.045$). Compared with public schools, private schools and other schools had significantly lower MMR rates ($\beta_3 = 0.375$, $p<0.001$; $\beta_2 = 0.279$, $p<0.001$), while schools with unknown type did not differ significantly ($\beta_4 = 0.335$, $p=0.106$). At the average level of state spending, the estimated marginal mean original MMR vaccination rates were 97.7% for public schools, 96.6% for other schools, 96.3% for unknown schools, and 96.1% for private schools. Substantial between-state variation remained after adjustment (state-level SD = 0.487).

3.4. Overall vaccination rates

The null model showed that 20.3% of the variation in overall vaccination rates was attributable to state-level differences (ICC = 0.203). After adjustment for spending and school type, the ICC was 0.199. State education spending was not significantly associated with overall vaccination rates ($\beta_1 = -0.139$, $p= 0.097$). At the average level of state spending, the estimated marginal mean original overall vaccination rates were 95.1% for public schools, 94.7% for other schools, 95.1% for unknown schools, and 93.7% for private schools. Compared with public schools, private schools had lower overall vaccination rates ($\beta_3 = 0.219$, $p < 0.001$), whereas “other” and unknown school types did not differ significantly ($\beta_2 = 0.061$, $p= 0.164$; $\beta_4 = -0.00$, $p= 1$). Between-state variation remained after adjustment (state-level SD = 0.445).

Table 2: Random Intercepts Linear Mixed Model Estimates for Log-Reflected MMR and Overall Vaccination Rates

Variable	MMR Vaccination Rate			Overall Vaccination Rate		
	Beta (SE)	95% CI	p-value	Beta (SE)	95% CI	p-value
Descriptive Statistics: Mean (SD) [Min, Max]						
Vaccination Rate	93.9 (9.4)	[1.0, 100.0]		90.9 (13.7)	[1.0, 100.0]	
Original Spending per Student (in \$1,000s)	11036.7 (3350.4)	[5885.1, 18999.5]		11036.7 (3350.4)	[5885.1, 18999.5]	
Fixed Effects						
Intercept	1.207 (0.173)	[0.854, 1.560]	< 0.001	1.774 (0.158)	[1.451, 2.097]	< 0.001
Education Spending Per Student (Scaled)	-0.185 (0.088)	[-0.366, -0.004]	0.045	-0.139 (0.081)	[-0.304, 0.027]	0.097
School Type: Other	0.279 (0.045)	[0.191, 0.366]	< 0.001	0.061 (0.044)	[-0.025, 0.148]	0.164
School Type: Private	0.375 (0.016)	[0.344, 0.406]	< 0.001	0.219 (0.016)	[0.188, 0.249]	< 0.001
School Type: Unknown	0.335 (0.201)	[-0.075, 0.746]	0.106	-0.000 (0.184)	[-0.375, 0.375]	1.000
Random Effects						
Random Effect: State (SD)	0.487			0.445		
Residual (SD)	0.904			0.893		
Model Fit & Validation						
Observations (Schools)	46213			46213		
Number of States	32			32		
ICC (Adjusted)	0.225			0.199		

Note: Modeled outcomes were heavily left-skewed and transformed using $\log((\text{Max} + 1) - \text{Rate})$. Consequently, a negative Beta indicates an INCREASE in the original vaccination rate. Education Spending Per Student was standardized (mean = 0, SD = 1); the Beta represents the effect of a one standard deviation increase. The reference category for School Type is 'Public'.

3.5. Sensitivity analyses and missing-data analyses

To assess the robustness of the primary findings, stratified analyses were conducted by school type and cohort. When the MMR model was stratified by school type, education spending remained significantly associated with MMR vaccination rates only among public schools ($\beta_{\text{public}} = -0.234$, $p = 0.027$), but not among private schools ($\beta_{\text{private}} = -0.12$, $p = 0.341$) or "other" schools ($\beta_{\text{other}} = -0.337$, $p = 0.081$). For overall vaccination rates, spending was not significant in any school-type subgroup ($\beta_{\text{public}} = -0.095$, $p = 0.2$; $\beta_{\text{private}} = -0.137$, $p = 0.172$; $\beta_{\text{other}} = -0.11$, $p = 0.133$). When stratified by cohort, spending was not significantly associated with MMR vaccination rates in either the 2017-2018 cohort ($\beta_{\text{spending scaled}} = -0.140$, $p = 0.183$) or the 2018-2019 cohort ($\beta_{\text{spending scaled}} = -0.109$, $p = 0.417$).

To further assess potential bias from missing data, mixed effects logistic regression models were fitted for missingness in MMR and overall vaccination outcomes. After accounting for state-level clustering, education spending was not a significant predictor of missingness for either outcome, whereas school type remained associated with missingness and state-level random-effect variances were large.

3.6. Model diagnostics

Figure 3 shows the diagnostic plots of the MMR model (3A and 3B) and the overall model (3C and 3D). The adjusted intraclass correlation coefficients supported the use of multilevel modelling, with 22.5% of the variance in MMR vaccination rates and 19.9% of the variance in overall vaccination rates attributable to between-state differences. After applying the log-reflected transformation, the Q-Q plots for both models showed approximate normality of residuals, although some deviation from the reference line remained in the tails. Fitted-versus-residuals plots did not show a clear funnel pattern. Variance inflation factors were low for all predictors ($\text{VIF} = 1.01$), indicating no evidence of problematic multicollinearity between state education spending and school type.

Figure 3A: Fitted values VS Residuals for MMR Vaccination Rate Model

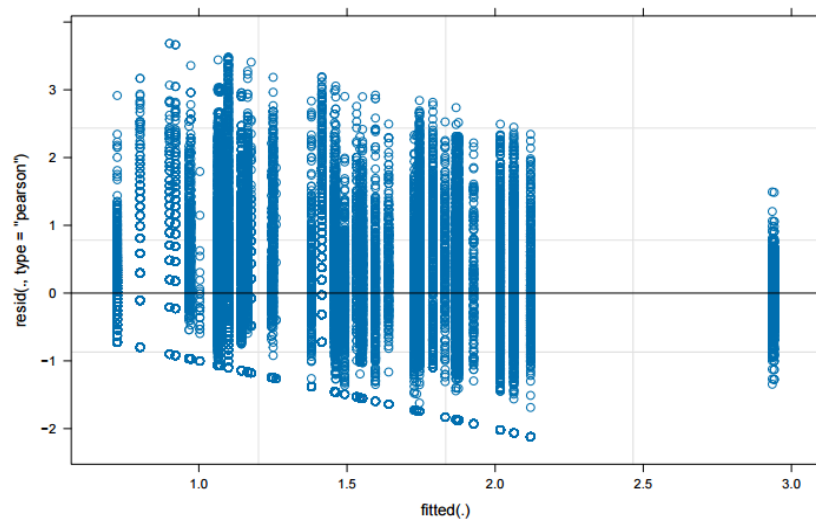


Figure 3B: Normal Q-Q plot for MMR Vaccination Rate Model

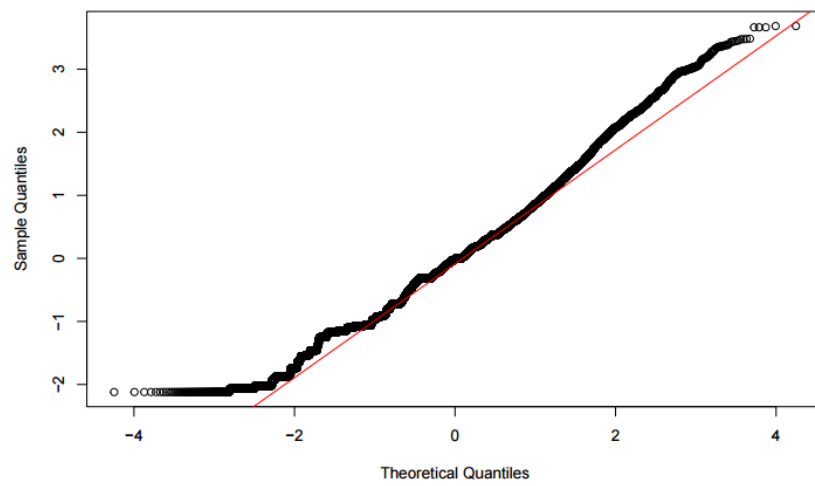


Figure 3C: Fitted values VS Residuals for Overall Vaccination Rate Model

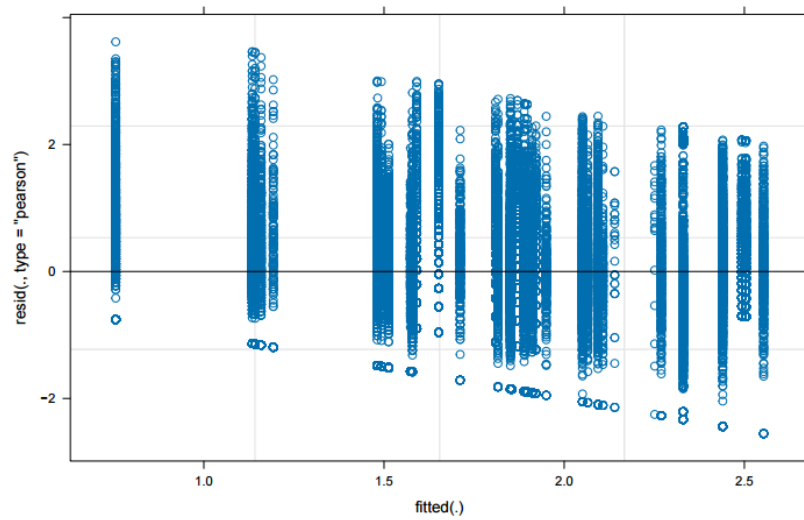
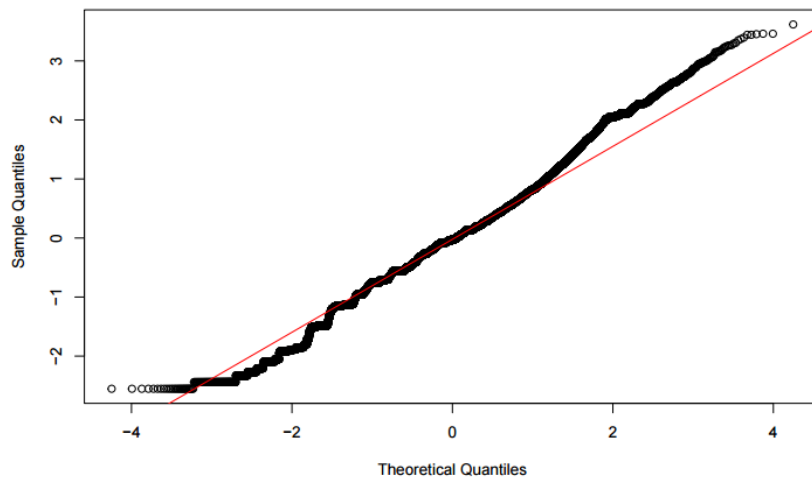


Figure 3D: Normal Q-Q plot for Overall Vaccination Rate Model



Note. Diagnostic plots for the final MMR (A and B) and overall (C and D) vaccination rates random intercepts models. Normality of residuals is confirmed by the Q-Q plot, and the residual plot shows no evidence of non-linearity or heteroscedasticity across the range of fitted values

4. Discussion

This study examined whether state-level public education spending was associated with school vaccination rates in the United States using data from 46,213 schools across all 32 states. Missing data were addressed using multiple imputation under a missing at random assumption, and random intercepts linear mixed models were used to account for clustering of schools within states.

4.1. Summary of findings

Higher state education spending per student was significantly associated with higher MMR vaccination rates, but not with overall vaccination rates. This suggests that while increased funding may improve compliance for specific, highly-monitored vaccines like MMR, it does not necessarily elevate the total immunization profile across all required school vaccines. At the average level of state spending, public schools had the highest estimated MMR rate (97.7%) and private schools the lowest (96.1%).

Stratified analyses suggested that this association was driven primarily by public schools. Among public schools, higher state spending was significantly associated with higher MMR rates, whereas no significant association was observed among private schools or other schools. One possible explanation is that public schools are more directly shaped by state funding structures, reporting requirements, and public policy than private or other school types.

By contrast, state spending was not significantly associated with overall vaccination rates. For overall rates, public schools had the highest estimated rate (95.1%), followed by other schools (94.7%), unknown schools (95.1%), and private schools (93.7%). Only private schools differed significantly from public schools. This difference between MMR and overall vaccination rates may reflect differences in reporting practices, public attention, or policy emphasis, since MMR vaccination is required for school entry in all states, whereas the overall measure includes a broader set of vaccines.

School type was also an important predictor across models. Private schools had significantly lower MMR vaccination rates than public schools, consistent with prior literature reporting higher exemption rates in private schools (Richards et al., 2013; Wang et al., 2014). In addition, a substantial proportion of the remaining variance was attributable to between-state differences (adjusted ICC = 22.5% for MMR and 19.9% for overall), indicating that unmeasured state-level factors, such as immunization laws, exemption policies (Omer et al., 2012), and public health infrastructure (Olive et al., 2018), likely continue to influence vaccination rates. Schools with unknown school type did not differ significantly from public schools in either model.

4.2. Limitations

Several limitations should be considered. First, the missingness was substantial. Although multiple imputation improved on complete-case analysis, the validity of the imputed results depends on the missing at random assumption, which cannot be fully verified. In particular, ten states had no observed MMR data, so results for that outcome depend partly on model-based reconstruction. Second, the cross-sectional design precludes causal inference. The observed association between state spending and MMR vaccination rates should therefore be interpreted as associative rather than causal, since both may reflect broader state-level commitments to child welfare or other unmeasured characteristics. Third, unmeasured confounding remains a concern. Important factors such as state immunization policies, parental attitudes, socioeconomic context, and school-level funding were not available in the dataset. The persistence of substantial between-state variation after adjustment supports the view that such unmeasured factors are important. Fourth, the significant spending effect for MMR was not robust across all sensitivity analyses. When models were stratified by cohort, the spending coefficient was not statistically significant in either the 2017-2018 or 2018-2019 subgroup, suggesting that the main finding may be sensitive to model specification or influenced by the imputation process. Fifth, state-level spending is only a coarse proxy for the resources available to individual schools. Because school-level funding data were unavailable, the analysis cannot determine whether increases in funding at the school level would directly improve vaccination rates. Finally, although the log-reflection transformation improved model fit, interpretation of coefficients is less intuitive, and residual diagnostics still showed some departure from normality in the tails.

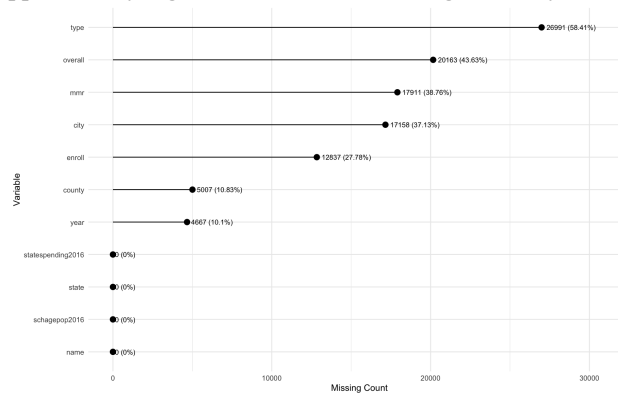
4.3. Conclusion

In this study, higher state education spending per student was significantly associated with higher school-level MMR vaccination rates, but not with overall vaccination rates. This association was observed primarily among public schools, suggesting that state investments in education may align more closely with vaccination outcomes within the public school system than in private or other school settings. Private schools consistently showed lower vaccination rates than public schools, highlighting a possible need for more targeted public health efforts in those settings.

Overall, the findings suggest that differences in school vaccination rates may reflect broader state-level contexts beyond education spending alone, including immunization policy, exemption laws, and public health infrastructure. Future research incorporating state policy measures, school-level funding data, and longitudinal designs would help clarify these relationships.

Appendix

Supplementary Figure S1: Number of Missing Values by Variables



Note. Type refers to school type, overall to overall vaccination rates, enroll to total number of students enrolled in the school, schagepop2016 is the school aged population of the state in 2016, name is the name of the school.

References

Omer, S. B., Richards, J. L., Ward, M., & Bednarczyk, R. A. (2012). Vaccination policies and rates of exemption from immunization, 2005–2011. *New England Journal of Medicine*, 367(12), 1170-1171.

Olive, J. K., Hotez, P. J., Damania, A., & Nolan, M. S. (2018). The state of the antivaccine movement in the United States: A focused examination of nonmedical exemptions in states and counties. *PLoS Medicine*, 15(6), e1002578.

Richards, J. L., Wagenaar, B. H., Van Otterloo, J., Gondalia, R., Atwell, J. E., Kleinbaum, D. G., Salmon, D. A., & Omer, S. B. (2013). Nonmedical exemptions to immunization requirements in California: A 16-year longitudinal analysis of trends and associated community factors. *Vaccine*, 31(29), 3009-3013.

Wang, E., Clymer, J., Davis-Hayes, C., & Buttenheim, A. (2014). Nonmedical exemptions from school immunization requirements: A systematic review. *American Journal of Public Health*, 104(11), e62-e84.